$\therefore \lceil x \rceil + \lceil y \rceil - \lceil x+y \rceil = 0$ or $1$ whenever $x$ and $y$ are real numbers. □

d) $\lceil xy \rceil = \lceil x \rceil \lceil y \rceil$ for all real numbers $x$ and $y$.

**Soln:** Let $x = 2.1, y = 1.2$

$\lceil xy \rceil = \lceil 2.1 \times 1.2 \rceil = \lceil 2.52 \rceil = 3$

$\lceil x \rceil \lceil y \rceil = \lceil 2.1 \rceil \lceil 1.2 \rceil = 3 \times 2 = 6$

$\therefore \lceil xy \rceil \neq \lceil x \rceil \lceil y \rceil$ for all real numbers $x$ and $y$. □

e) $\lceil x/2 \rceil = \left\lfloor \frac{x+1}{2} \right\rfloor$ for all real numbers $x$.

**Soln:** Let, $x = \frac{1}{2}$ $\therefore \lceil x/2 \rceil = \lceil (1/2)/2 \rceil = \lceil 1/4 \rceil = 1$

$\left\lfloor \frac{x+1}{2} \right\rfloor = \left\lfloor \frac{\frac{1}{2}+1}{2} \right\rfloor = \left\lfloor \frac{3}{4} \right\rfloor = 0.$

$\therefore \lceil x/2 \rceil \neq \left\lfloor \frac{x+1}{2} \right\rfloor$ for all real numbers $x$. □

76) Prove or disprove each of these statements about the floor and ceiling functions.

a) $\lfloor \lceil x \rceil \rfloor = \lceil x \rceil$ for all real numbers $x$.

**Soln:** By Defn. 12, $\lceil x \rceil$ is the smallest integer larger than or equal to $x$.

By Defn. 12, $\lfloor \lceil x \rceil \rfloor$ is the largest integer smaller than or equal to $\lceil x \rceil$.

$\therefore \lceil x \rceil \in \mathbb{Z}$, $\lfloor \lceil x \rceil \rfloor = \lceil x \rceil$, $\forall x \in \mathbb{R}$. □

b) $\lfloor x+y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor$ for all real numbers $x$ and $y$.

**Soln:** Let, $x = 2.9, y = 1.2$

$\lfloor x+y \rfloor = \lfloor 2.9+1.2 \rfloor = \lfloor 4.1 \rfloor = 4$

$\lfloor x \rfloor + \lfloor y \rfloor = \lfloor 2.9 \rfloor + \lfloor 1.2 \rfloor = 2+1 = 3$

$\lfloor x+y \rfloor \neq \lfloor x \rfloor + \lfloor y \rfloor$ for all real numbers $x$ and $y$. □

c) $\lceil \lceil x/2 \rceil /2 \rceil = \lceil x/4 \rceil$ for all real numbers $x$

**Soln:** Let, $\lceil x/2 \rceil = m$. $\therefore x/2 + \varepsilon_1 = m$, where $0 \leq \varepsilon_1 < 1$.

Let, $\lceil m/2 \rceil = n$. $\therefore m/2 + \varepsilon_2 = n$, where $0 \leq \varepsilon_2 < 1$. $\Rightarrow$

$\Rightarrow (x/2 + \varepsilon_1)/2 + \varepsilon_2 = n \Rightarrow x/4 + \varepsilon_1/2 + \varepsilon_2 = n$

$\therefore 0 \leq \varepsilon_1 < 1 \Rightarrow 0 \leq \varepsilon_1/2 < 1/2$

By Defn. 12, $m$ must be an integer.